Kubernetes Worker Node Configuration Guide

A step-by-step setup reference for configuring and joining a worker node to an existing Kubernetes cluster using **kubeadm** on Ubuntu/Debian.

Step 1: System Prerequisites

Turn off swap memory (required by Kubernetes) and configure network bridging parameters.

```
# 1. Disable swap
sudo swapoff -a
sudo sed -i '/ swap / s/^(.*)$/#\1/g' /etc/fstab

# 2. Forward IPv4 and configure sysctl
cat <<EOF | sudo tee /etc/modules-load.d/k8s.conf
overlay
br_netfilter
EOF

sudo modprobe overlay
sudo modprobe br_netfilter

cat <<EOF | sudo tee /etc/sysctl.d/k8s.conf
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
EOF

sudo sysctl --system
```

Step 2: Install Container Runtime (containerd)

Install containerd and update the configuration to utilize SystemdCgroup.

```
sudo apt-get update && sudo apt-get install -y containerd
sudo mkdir -p /etc/containerd
containerd config default | sudo tee /etc/containerd/config.toml
sudo sed -i 's/SystemdCgroup = false/SystemdCgroup = true/g'
/etc/containerd/config.toml
sudo systemctl restart containerd
sudo systemctl enable containerd
```

Step 3: Install Kubeadm, Kubelet, and Kubectl

Add the official Kubernetes repository and install the standard binaries.

```
sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.30/deb/Release.key | sudo gpg
```

```
--dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg]
https://pkgs.k8s.io/core:/stable:/v1.30/deb/ /' | sudo tee
/etc/apt/sources.list.d/kubernetes.list
sudo apt-get update
sudo apt-get install -y kubelet kubeadm kubectl
sudo apt-mark hold kubelet kubeadm kubectl
```

Step 4: Join the Node to the Cluster

Execute the join command generated during the master node initialization to register the worker node.

```
sudo kubeadm join <MASTER_IP>:6443 --token <TOKEN> \
--discovery-token-ca-cert-hash sha256:<HASH>
```

Step 5: Verify Node Status

Run the following check on the **Master Node** to confirm connectivity:

```
kubectl get nodes
```